

MFR Data Analytics Program

Final Project Student Guide

Module 9 — Exploratory Data Analysis (EDA)

What The Project Demonstrates

By completing this project you will show that you can:

Module skill	What you demonstrate
Python & Pandas	Load, explore and manipulate a real dataset using pandas and numpy
Data Cleaning	Identify and handle missing values, outliers, type errors and inconsistencies
Statistics	Apply descriptive statistics and, where relevant, hypothesis testing
Visualisation	Create clear, well-labelled charts using matplotlib, seaborn and/or plotly
EDA	Use histograms, boxplots, pairplots, heatmaps, time series and correlation analysis
Storytelling	Communicate your findings in a structured, narrative format

Choosing Your Dataset

You are free to choose any dataset that meets the following requirements:

- At least 5,000 rows and 8 columns
- Contains at least one date/time column
- Contains at least one categorical and one numerical variable
- Is publicly available and does not contain personal or sensitive data
- Has not been used in any of the course mini-projects (no Superstore, no coffee shop, no HR data)

Suggested sources

Kaggle (kaggle.com/datasets) · Our World in Data (ourworldindata.org) · UCI ML Repository · Google Dataset Search · data.gov · Any open government data portal

Not sure which dataset to choose? Ask your mentor for approval before starting. A dataset that is too clean or too small will limit the depth of your analysis.

Project Structure

Your Jupyter Notebook must be structured in the following six sections. Each section maps directly to skills learned in the course.

Section 1: Dataset Introduction & Initial Exploration — Modules 2 & 3

Tasks:

- Describe the dataset: what it contains, where it comes from, and why it is interesting
- Load the data with pandas and display the first rows
- Check shape, data types, and column names
- Answer: How many rows? How many columns? What does each column represent?
- Check for duplicates and summarise with `.describe()`

Deliverable:

A written introduction (markdown cell) + code cells with initial exploration output

Section 2: Data Cleaning & Preparation — Module 4

Tasks:

- Identify and handle missing values (document your decisions)
- Convert columns to the correct data types (especially dates)
- Handle outliers: detect them, decide whether to remove or keep, and justify
- Clean any text columns if needed (strip spaces, standardise categories)
- Create at least one new feature through feature engineering

Deliverable:

A clean, ready-to-analyse DataFrame with a written summary of all decisions made

Section 3: Descriptive Statistics — Module 3

Tasks:

- Compute central tendency and dispersion for all key numerical variables
- Compare group statistics using `groupby`
- Identify and comment on variables that are skewed or have high variance
- Formulate at least 3 analytical questions you will investigate in the next sections

Deliverable:

A statistics summary table + 3 clearly stated analytical questions

Section 4: Univariate & Multivariate Visual Exploration — Modules 5 & 9**Tasks:**

- Plot histograms for all key numerical variables — interpret the shape
- Use boxplots to compare distributions across groups
- Build a pairplot and a correlation heatmap (Pearson and/or Spearman)
- Choose at least 5 chart types from the course (bar, scatter, line, box, heatmap, etc.)
- Each chart must have a title, axis labels, and a written interpretation below it

Deliverable:

At least 8 visualisations, each followed by a written interpretation

Section 5: Time Series & Correlation Analysis — Module 9**Tasks:**

- Resample the data to at least two time frequencies (e.g. monthly and quarterly)
- Plot a rolling average to reveal trend and seasonality
- Identify and describe at least one seasonal pattern
- Conduct a correlation analysis: Pearson vs Spearman, justify your choice
- Plot a ranked correlation bar chart showing what drives your main variable of interest

Deliverable:

Time series charts + correlation analysis + written narrative

Section 6: Conclusions & Business Recommendations**Tasks:**

- Summarise your 3–5 most important findings
 - Explain what surprised you and why
 - Provide 2/3 concrete recommendations based on your analysis
 - Identify the limitations of your analysis and what you would explore with more time
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